University of Westminster

Database Systems

5COSC020C

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Abstract.

It is all about this report concerning the way I designed and created conceptual and logical database models of two business situations, namely, Foodz@Yourz and BoilHeaterz. I have been keen on applying the concepts of relational database design using UML notations in order to make the systems robust, should be scaled, and have to be logically consistent.

In the case of Foodz@Yourz, I have created an Enhanced Entity Relationship Diagram (EERD) to design multi-store ordering and delivery aspects of the company in form of a grocery. I also assembled tabular data dictionaries and maintained a decision log to record all the decisions that we made.

The second section of the coursework required me to translate one conceptual model of BoilHeaterz into a Logical Entity Relationship Diagram (ERD). I needed to work out the relationships, specialisations, and multiplicities to ensure that everything is in line with relational databases.

In general, it was aimed to demonstrate that I could transform business requirements, then to data models, and to provide the justification of every design choice with a strong rationale on the basis of data theory and the UML standards.

Acknowledgment.

I would like to thank our module leader in charge of 5COSC020C: Database Systems, Ms.Janani Harischandra, for guiding me throughout this project. And next, I would like to pay my gratitude to Mr. Pubudu Arachchige and Ms. Ayeshka Jayasundara for their unwavering support, encouragement, and guidance throughout this coursework.

Moreover, I am grateful for the support of the internet resources such as documentation and academic forums, and various online learning platforms, which helped me improve and understand database modeling concepts and problem-solving.

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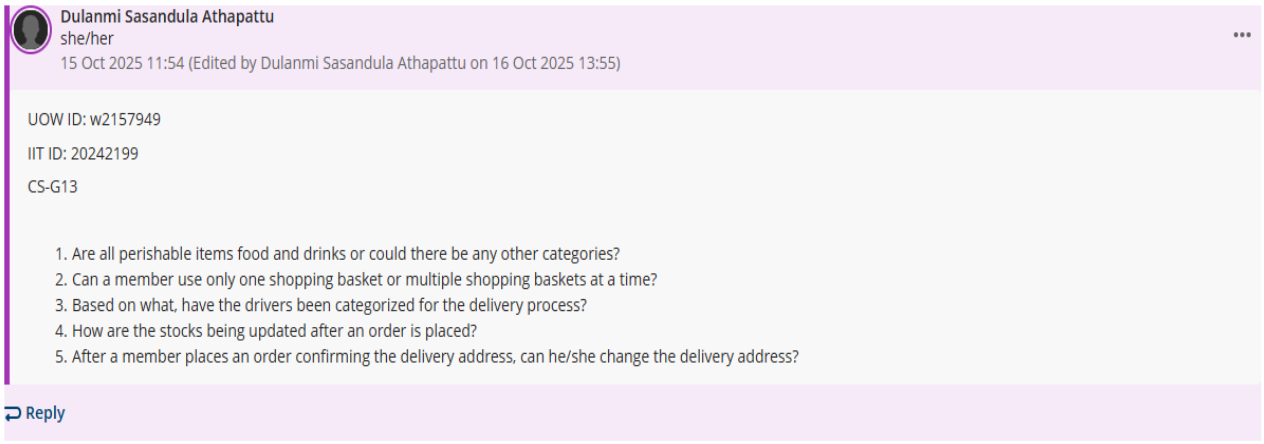
Business Questions


Dulanmi Sasandula Athapattu
she/her
15 Oct 2025 11:54 (Edited by Dulanmi Sasandula Athapattu on 16 Oct 2025 13:55)

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CS-G13

1. Are all perishable items food and drinks or could there be any other categories?
2. Can a member use only one shopping basket or multiple shopping baskets at a time?
3. Based on what, have the drivers been categorized for the delivery process?
4. How are the stocks being updated after an order is placed?
5. After a member places an order confirming the delivery address, can he/she change the delivery address?

Reply

Figure 1 Business Questions

Answers:

1. All perishable items are not food products and drink products. There can be other perishable categories too [ex: - toiletries].

2. A member can have multiple named baskets which they can create, update and keep more than one basket at a time.

3. Drivers are categorized by their role and their association with vehicles for deliveries during specific time windows.

4. The system only needs to record the stock level for each product at each store. Stock management is handled only by the stock level for a single product at a single store. Recording changes to product stock levels are out of scope.

5. Once an order is placed by confirming the delivery address, the address cannot be changed. The application handles the order editing and cancellation process; the database only records the order status.

PART – A

Conceptual EERD

Enhanced Entity Relationship Diagram [EERD]

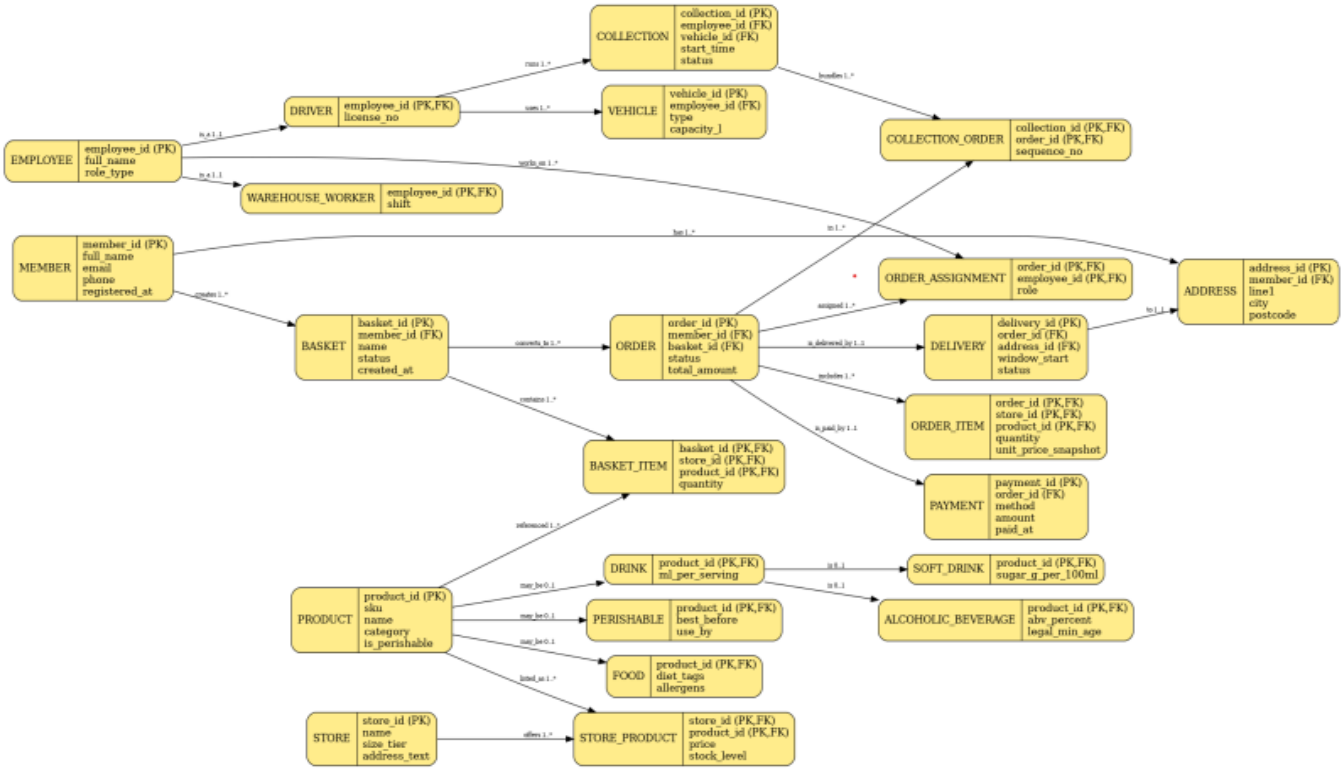

Figure 2 AI-generated EERD

Question 01: CRITICAL EVALUATION

Strengths	Justification
Recognition of the main entity domains.	Here, the main entities have been clearly identified [example: Member, Store, Product, Order, Employee, Payment, etc.], which also provides a conceptual foundation for the EERD.
Includes specialisations for the main entity "Product".	Here, specialization is used for the product subtype [example: drinks have been divided into alcoholic vs soft drinks], which basically helps to organize data and avoid repetition and keeps the diagram tidy.
The EERD diagram correctly separates the order and payment entities and has connected it correctly.	Here, each order has been linked to a payment, as a payment must be made after every order is placed, before the delivery, and also manages the payment method and keeps the data organized along with the transaction process.

Weaknesses	Justification
Multiplicities were missing.	For example, Store-product should be many-to-many because products can appear in multiple stores. Missing multiplicities reduce accuracy.
Perishable products haven't been specialized.	Here, the product entity has been specialized into perishable, drink, and food. It seems to be a weakness, as perishables should be specialized into food and drink, not the product.
Doubtful handling of the shopping basket and order.	Here, the basket and order should be separated without connecting, as the basket can be editable before purchase, but the order is final after the payment.

Question 02: DECISION LOG

Modelling Decision	Justification
Added a specialization for vehicle; Two-wheeler and Four-wheeler.	Drivers can use two types of vehicles which means that Two-wheelers handle small collections and Four-wheelers handle large collections.
Created a specialization for Product: Perishable and Non-perishable.	Products are supposed to be specialized as perishable and non-perishable before they are further specialized.
Created a specialization for perishable; Food and Drinks.	Perishable products can be food and drinks as they have unique data like the expiry date.
Created a specialization for Employees; Driver and Warehouse Worker.	So the business has two employees who are Drivers who handle the deliveries and Warehouse Workers who handle the orders. So, specializing it clearly figures out their distinct responsibilities.
Defined a one-to-many relationship between Member and Basket.	Each member can create multiple baskets, but each basket belongs to only one unique member.
Using of derived attributes.	Derived attributes were used in order to represent values that are been calculated from other data. total Amount, dateOfBirth like attributes have been represented as derived attributes.
Design consistency.	Only entities, attributes along with primary keys and relationships are represented in order to maintain the clarity of the database design.
Use of attributes.	Only the essential attributes are been included in order to keep the EERD simple and avoid other issues including technical issues.
Added ternary relationships between Basket, Store and Product.	Here, it links Basket, Store and Product to represent that the basket contains specific products from specific stores, where the price and the stock depend on the relevant store.
Added ternary relationship between Order, Product and Collection.	Represents each collection includes specific products and orders, allowing accurate tracking of which products are gathered for each collection.

Question 03: ENTITIES

Entity Name	Brief Explanation
w2157949_Employee	Represents all the staff members at Foodz@Yourz.
w2157949_Product	Stores all the items' details, including perishable and non-perishable items, prices, and stock levels for sales across multiple stores.
w2157949_Basket	A virtual container used to hold selected products before placing an order.
w2157949_Member	Stores the information of the registered customers who purchase and receive the groceries.
w2157949_Payment	Record the details of payments, including the payment type and the transaction information, made by the customers.
w2157949_Order	Represents the purchase confirmation connecting with the selected product, payment, and delivery.
w2157949_Delivery	Stores the information such as the delivery date, address, and assigned driver, relevant for the delivery process for each order.
w2157949_Collection	Represents the confirmed orders picked up from the stores or warehouse by a driver before the delivery.
w2157949_Store	Stores the location and product offerings of the partnered grocery stores of Foodz@Yourz.
w2157949_Vehicle	Represents the type of vehicle used for deliveries.

Question 04: SPECIALISATIONS

General Entity	Specialised Entity	Brief Explanation
w2157949_Products	w2157949_Perishable	Represents the products that are about to expire.
w2157949_Products	w2157949_NonPerishable	Represents the products that lasts long without spoiling.
w2157949_Perishable	w2157949_Food	Specialised entity of perishables that includes food items that are about to expire.
w2157949_Perishable	w2157949_Drink	Specialised entity of perishables that includes beverages that are about to expire.
w2157949_Payment	w2157949_CardPayment	Represents the payments made using credit or debit cards.
w2157949_Payment	w2157949_MobilePayment	Represents the payments done using mobile apps or digital wallets.
w2157949_Drink	w2157949_SoftDrink	Represents non-alcoholic drinks such as juices, soda, etc.
w2157949_Drink	w2157949_AlcoholicBeverages	Represents the beverages containing alcohol.
w2157949_Vehicle	w2157949_TwoWheeler	Represents small vehicles used for delivery to cover small collections.
w2157949_Vehicle	w2157949_FourWheeler	Represents large vehicles used for delivery to cover larger collections.
w2157949_Employee	w2157949_Driver	Represents the employees who are responsible for collecting confirmed orders and delivering them to the members' home addresses.
w2157949_Employee	w2157949_WarehouseWorker	Represents the employees who are responsible for placing orders and ensuring that their processed properly.

Question 05: RELATIONSHIPS & MULTIPLICITIES

Entity Name	Multiplicity	Relationship	Multiplicity	Entity Name	Brief justifications for the multiplicity (4 statements for each relationship)
w2157949_Member	1..1	creates	1..*	w2157949_Basket	Member creates at least one basket. Member creates at most many baskets. A basket is created by at least one member. A basket is created by at most one member.
w2157949_Member	0..*	purchases	1..*	w2157949_Product	Member purchases at least one product. Member purchases at most many products. A product is been purchased by at least no members. A product is been purchased by at most many members.
w2157949_Basket	1..*	selects	1..*	w2157949_Product	A basket has at least one product. A basket has at most many products. A product has been at least in one basket. A product has been at most in many baskets.
w2157949_Product	1..*	selects	1..*	w2157949_Store	A product has at least one store.

					A product has at most many stores. A store has at least one product. A store has at most many products.
w2157949_Basket	1..1	places	0..*	w2157949_Order	A basket places at least zero orders. A basket places at most many orders. A order is been placed by at least one basket. A order is been placed by at most one basket.
w2157949_Order	1..1	takes	1..1	w2157949_Payment	A order takes at least one payment. A order takes at most one payment. A payment is been taken by at least one order. A payment is been taken by at most one order
w2157949_Order	1..1	collects	1..*	w2157949_Product	An order collects at least one product. An order collects at most many products. A product is been collected by at least one order. A product is been collected by at most one order.

w2157949_Collection	1..*	collects	1..*	w2157949_Product	A collection collects at least one product. A collection collects at most many products. A product is been collected by at least one collection. A product is been collected by at most many collections.
w2157949_Driver	1..1	runs	1..*	w2157949_Collection	A driver runs at least one collection. A driver runs at most many collections. A collection is been run by at least one driver. A collection is been run by at most one driver.
w2157949_Driver	1..*	uses	1..*	w2157949_Vehicle	A driver uses at least one vehicle. A driver uses at most many vehicles. A vehicle is been used by at least one driver. A vehicle is been used by at most many drivers.
w2157949_Warehouse Worker	1..1	prepares	1..*	w2157949_order	A warehouse worker prepares at least one order. A warehouse worker prepares at most many orders. An order is been prepared by at least

					one warehouse worker. An order is been prepared by at most one warehouse workers.
w2157949_Delivery	1..*	collects	1..1	w2157949_Collection	A delivery collects at least one collection. A delivery collects at most one collection. A collection is been collected by at least one delivery. A collection is been collected by at least many deliveries.
w2157949_Order	1..1	generates	1..1	w2157949_Delivery	An order is generated by at least one delivery. An order is generated by at most one delivery. A delivery is been generated by at least one order. A delivery is been generated by at most one order.
w2157949_Collection	1..*	collects	1..1	w2157949_Order	A collection collects at least one order. A collection collects at most one order. An order is been collected by at least one collection. An order is been collected by at most many collections.

Question 06: ATTRIBUTES & PRIMARY KEYS

Entity Name	Attributes for each entity (include PK)	Brief Explanation
w2157949_Member	w2157949_memberID {PK} w2157949_memberName w2157949_email w2157949_address w2157949_phone w2157949_dateOfBirth	Details of the registered customers.
w2157949_Product	w2157949_productID {PK} w2157949_productName w2157949_description w2157949_category w2157949_price w2157949_productIngredients	Information about the products in the store.
w2157949_Basket	w2157949_basketID {PK} w2157949_totalAmount w2157949_basketTimeStamp w2157949_basketStatus w2157949_memberID w2157949_productID	A temporary basket created by the registered members.
w2157949_Employee	w2157949_employeeID {PK} w2157949_employeeName w2157949_empContactNo w2157949_roleType	Staff entities of the business.
w2157949_Payment	w2157949_paymentID {PK} w2157949_paymentType w2157949_paymentTimeStamp w2157949_paidAmount w2157949_paymentReference w2157949_paymentStatus w2157949_orderID	Payment entries of each and every order.
w2157949_Order	w2157949_orderID {PK} w2157949_orderDate w2157949_orderRef w2157949_orderTimeStamp w2157949_orderStatus w2157949_totalAmount w2157949_basketID	Confirmed order from a generated basket.

w2157949_Delivery	w2157949_deliveryID {PK} w2157949_deliveryDate w2157949_driverName w2157949_deliveryRef w2157949_delTimeStamp w2157949_delStatus w2157949_address w2157949_orderID	Representing the distribution process of the confirmed orders.
w2157949_Collection	w2157949_collectionID {PK} w2157949_timeStamp w2157949_RefNumber w2157949_cStatus w2157949_collectionType w2157949_employeeID w2157949_orderID	Group multiple orders at a time for dispatching.
w2157949_Store	w2157949_storeID {PK} w2157949_storeName w2157949_location w2157949_stockLevel w2157949_productID	Stores in partnership within the system.
w2157949_Vehicle	w2157949_vehicleID {PK} w2157949_vehicleType w2157949_licenseNo w2157949_capacity w2157949_manufactureDetails	Details of the vehicles used for delivery.
w2157949_Driver	w2157949_employeeID w2157949_vehicleID w2157949_licenseNo	Subclass of employees who handles deliveries.
w2157949_WarehouseWorker	w2157949_employeeID w2157949_shift	Subclass of employees who manage the orders.
w2157949_Perishable	w2157949_productID w2157949_storageType w2157949_nutritionalInfo w2157949_allergenInfo w2157949_validityInfo	Subtype of perishable foods.
w2157949_NonPerishable	w2157949_productID w2157949_storageType	Subclass of products.
w2157949_Food	w2157949_productID w2157949_foodCategory	Subclass of perishable food items.
w2157949_Drink	w2157949_productID w2157949_volume	Subclass of perishable drinks.

w2157949_SoftDrink	w2157949_productID w2157949_flavour	Subclass of non-alcoholic drinks.
w2157949_Alcoholic	w2157949_productID w2157949_alcoholicPercentage	Subclass of alcoholic drinks.
w2157949_CardPayment	w2157949_paymentID w2157949_totalAmount	Subclass of payments done using cards.
w2157949_MobilePayment	w2157949_paymentID w2157949_totalAmount	Subclass of payments done using mobile apps.
w2157949_TwoWheeler	w2157949_vehicleID w2157949_collectionID	Subclass of vehicles used to do small deliveries.
w2157949_FourWheeler	w2157949_vehicleID w2157949_collectionID	Subclass of vehicles used to do big deliveries.

Question 07: QUALITY & NOTATIONS OF CONCEPTUAL EERD

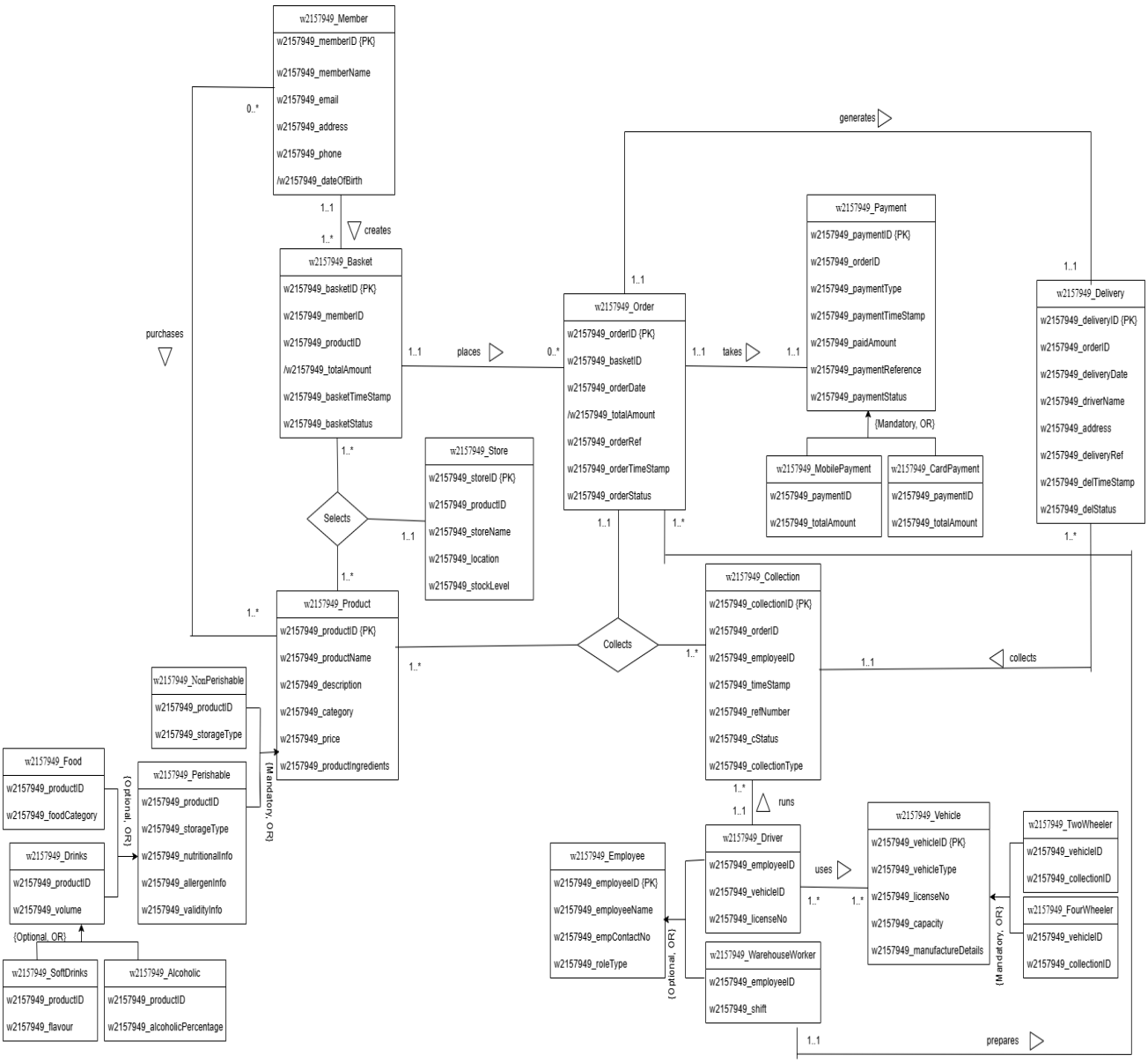

Figure 3 Conceptual EERD Diagram [Part – A]

PART – B

Conceptual Enchanced Entity Relationship Diagram [EERD]

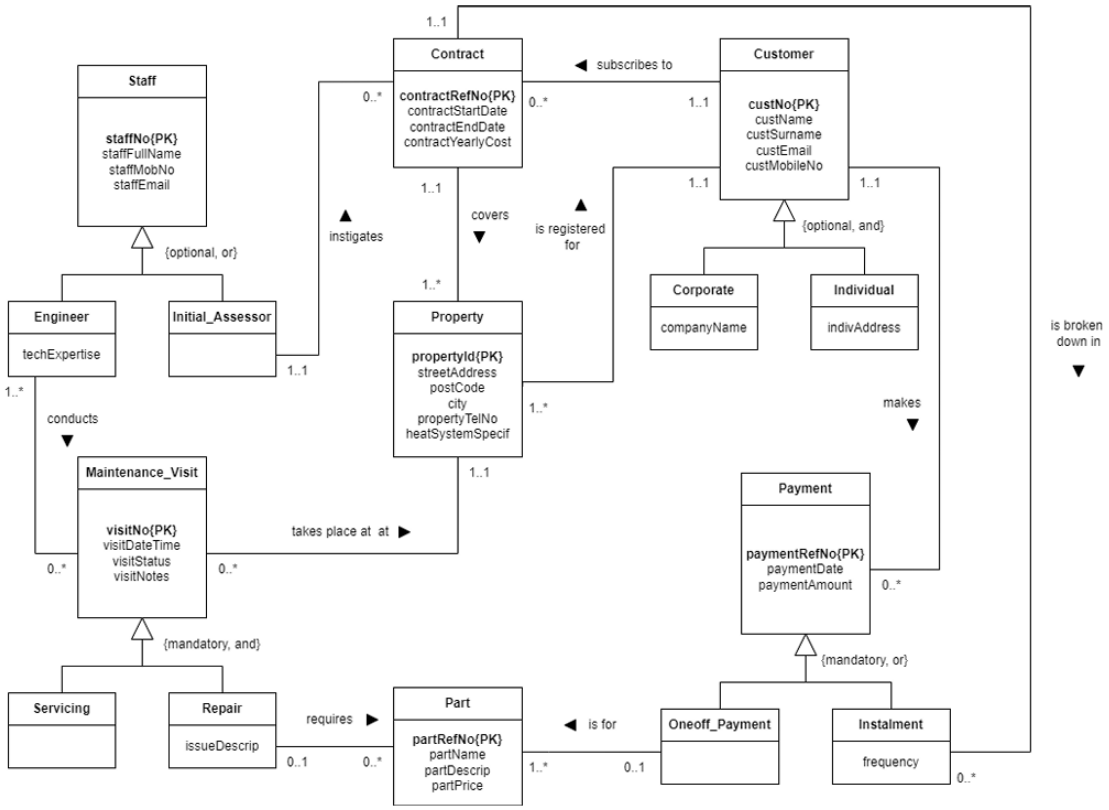

Figure 4 Conceptual EERD [Part - B]

Question 08: MAPPING CONCEPTUAL EERD TO LOGICAL ERD

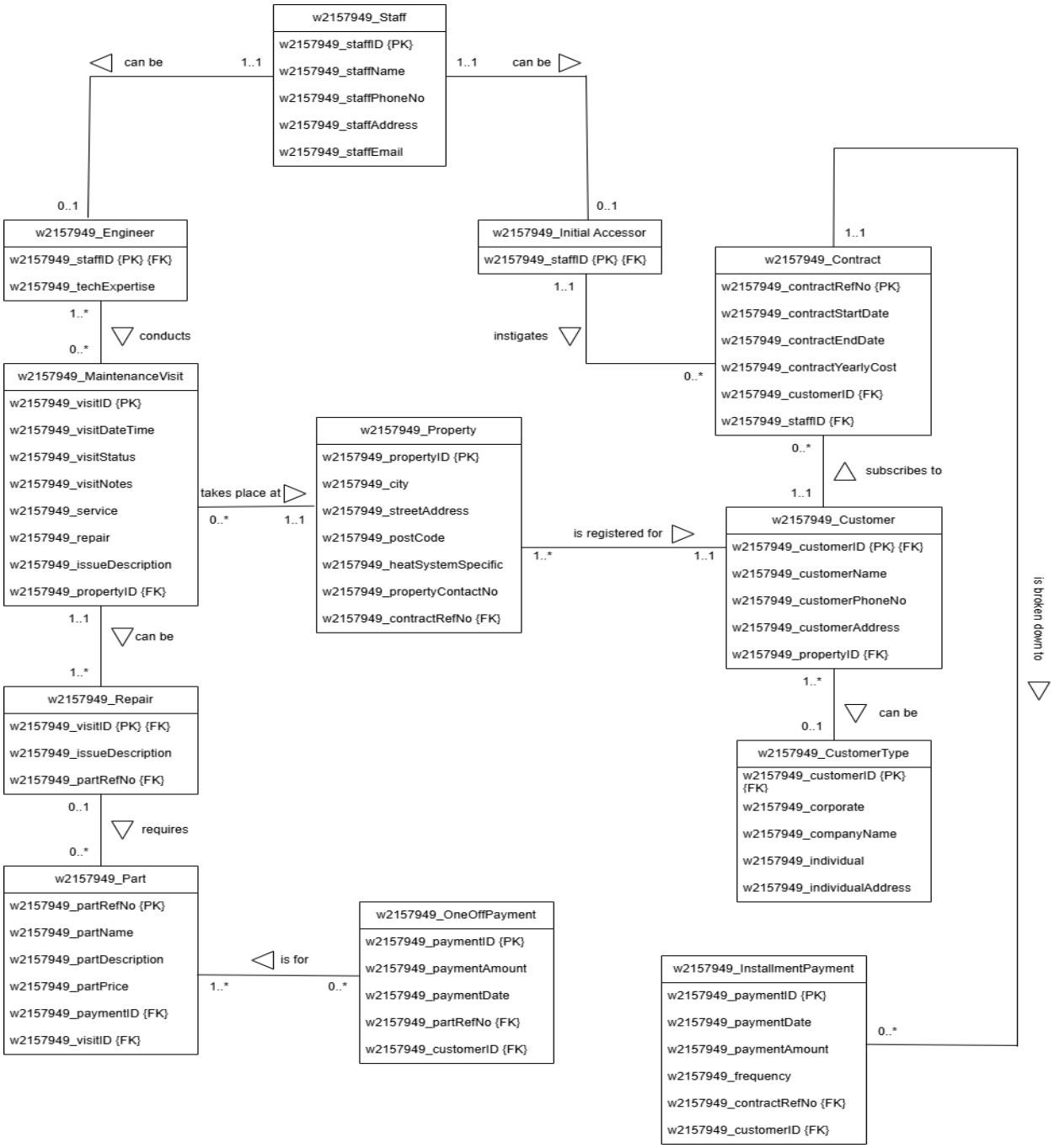

Figure 5 Logical EERD

Question 09: MAPPING LOG

Mapping Decisions	Explanation
Specialisation Mapping of Staff in Initial Assessor and Engineer	The Staff entity further subdivides into two optional sub-types Engineer and Initial Assessor. The one table per sub-type rule will be used to create a table per sub-type and the primary as well as the foreign key will be the staffID in order to ensure that the integrity is maintained as well as the optional OR relationship is exhibited.
Specialisation of Customer-to-customer type	Customer entity may be either Individual or Corporate. It is just configured with a single CustomerType table with subtype attributes such as the companyName and individualAddress. The primary and foreign key of the customerID is used to demonstrate that optional AND relationship.
Specialisation of Maintenance visit to Repair and Servicing	In such a way, the MaintenanceVisit entity is divided into two obligatory sub entities Repair and Servicing. All of those sub-types have their own table, and the primary key and a foreign key of the table is visitID, which is just one of the necessary and sufficient conditions to ensure that all maintenance visits will be classified as either a repair or a service by the mandatory AND rule.
Payment specialization [One-off payment and Installments]	The OneOffPayment and InstallmentPayment tables inherit from Payment. Each uses paymentID as both the primary and foreign key, with extra attributes like frequency for installments. Foreign keys link to relevant entities based on payment type, following the mandatory OR constraint.
Strong entities were mapped as core base tables in the logical schema.	All the main entities have been mapped directly to the relational tables and their primary keys remained unchanged in order to uniquely identify each real-world entity and ensure relational consistency.

Conclusion.

I completed this coursework and it provided me with a decent understanding of the process of creating a database. I worked my way up and down the pyramid on ideas of high level to the cool back-end where the database actually executes. The EER diagram of the Foodz@Yourz which I had the opportunity to model made it possible to represent all the complexities of the real world (i.e., multiple specialties and store relationships, etc.) and have it work when one considers a real retail chain. After that using the BoilHeaterz logical model I nailed the mapping rules, to such that a relational schema is obtained that is fully normalized. Adhering to a decision-making flow and using the UML rules, the two models became incredibly accurate, transparent, and easy to maintain.

My experience on doing this project made me realize the importance of a cool design process. It was a type of challenge which forced me to be highly critical and maintain the documentation tight. All that effort is necessary to ensure that an actual database supports the way of how a business actually functions and keeps the information safe and stable.